Break Into AI With ZERO Experience

The 12-Week Plan

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No CS degree. No bootcamp. No $5,000 course.
Just 12 weeks of focused building.

WEEK 1-2: LEARN PYTHON

Don't overthink this. Python is the only language you need to break into AI. Learn just the basics — you don't need to master it before moving on.

What to learn:

- Variables, data types, strings
- If/else statements, loops (for, while)
- Functions and return values
- Lists, dictionaries, tuples
- Reading/writing files
- pip and installing packages

Free resources:

- **Kaggle Learn Python** — Interactive, beginner-friendly
  https://www.kaggle.com/learn/python
- **CS50 Introduction to Python** — Harvard's free course
  https://cs50.harvard.edu/python/
- **Corey Schafer Python Playlist** — Best YouTube tutorial
  https://www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU
- **Automate the Boring Stuff** — Free online book
  https://automatetheboringstuff.com/

Goal:

By end of Week 2: You can write a Python script that takes input, does something with it, and gives output. That's it.

WEEK 3: BUILD SOMETHING DUMB

I'm serious. You learn 10x faster by building than by watching tutorials. The project doesn't need to be impressive — it needs to EXIST.

Quick project ideas (pick one):

- A chatbot that roasts you based on your name
- A script that summarizes any webpage you paste in
- A random meal picker that texts you what to eat
- A quiz app that tests your Python knowledge
- A script that renames all files in a folder automatically

Tools you'll need:

- **OpenAI API** — Free $5 credit on signup
  https://platform.openai.com/
- **Streamlit** — Build a UI in Python in minutes
  https://streamlit.io/

How to get the OpenAI API key:

- Go to https://platform.openai.com/signup
- Sign up and add a payment method (you get free credits)
- Go to API Keys → Create new secret key
- Save it in a .env file: OPENAI_API_KEY=sk-your-key-here

Goal:

By end of Week 3: You have one working project that uses an API. Push it to GitHub.

WEEK 4-5: LEARN PROMPT ENGINEERING

This is the highest-paid skill in tech right now. Companies pay $120K-$180K for people who can write great prompts. You can learn this in 2 weeks — not 2 years.

What to learn:

- System prompts vs user prompts
- Zero-shot vs few-shot prompting
- Chain-of-thought prompting
- Temperature and token settings
- Output formatting (JSON mode, structured output)
- Role-based prompting

Free resources:

- **DeepLearning.AI Prompt Engineering**
  https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/
- **OpenAI Prompt Engineering Guide**
  https://platform.openai.com/docs/guides/prompt-engineering
- **Anthropic Prompt Engineering Guide**
  https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering
- **Brex Prompt Engineering (GitHub)**
  https://github.com/brexhq/prompt-engineering

Practice exercises:

- Summarize any article in exactly 3 bullet points
- Extract structured data (name, email, company) from messy text
- Generate a weekly meal plan as JSON
- Act as a code reviewer and find bugs
- Translate text while keeping the tone casual

Goal:

By end of Week 5: You can write prompts that consistently produce the exact output you want.

WEEK 6-8: BUILD A RAG APP

RAG (Retrieval-Augmented Generation) is the #1 thing companies hire AI engineers for. It's how you make AI answer questions about YOUR data — documents, PDFs, websites.

The project: PDF Q&A Chatbot

Upload any PDF → ask questions about it → get accurate answers with sources.

Tech stack:

- Python
- LangChain — https://langchain.com
- OpenAI API (for embeddings + chat)
- ChromaDB (free vector database) — https://www.trychroma.com
- Streamlit (for the UI)
- PyPDF for reading PDFs

Step-by-step:

- Load a PDF using PyPDF
- Split it into chunks (500-1000 characters each)
- Create embeddings using OpenAI's embedding model
- Store embeddings in ChromaDB
- When user asks a question, find the most relevant chunks
- Send those chunks + the question to GPT-4
- Display the answer with source references

Free resources:

- **LangChain RAG Tutorial**
  https://python.langchain.com/docs/tutorials/rag/
- **LangChain Academy** — Free comprehensive course
  https://academy.langchain.com/
- **Hugging Face RAG Course**
  https://huggingface.co/learn/cookbook/rag_with_hugging_face_gemma_mongodb

Goal:

By end of Week 8: You have a working RAG app deployed on Streamlit Cloud that anyone can use.

WEEK 9-10: DOCUMENT EVERYTHING ON GITHUB

Your GitHub is your resume. No recruiter cares about certificates. They care about what you've BUILT.

What your GitHub profile needs:

- A profile README (introduce yourself)
- 3-5 pinned repositories
- Green contribution graph (shows you're active)

What every project repo needs:

- Clear project title + one-line description
- Screenshot or demo GIF
- Tech stack badges
- How to install and run it
- What problem it solves
- requirements.txt with all dependencies
- .env.example file (never commit real API keys)
- A live demo link

Profile README template:

```
# Hi, I'm [Your Name]

AI Engineer | Building with LLMs, RAG, and LangChain
Based in [Your City]
Currently building: [your latest project]
Learning: [what you're learning now]

## My Projects
- Project 1 (link) — one-line description
- Project 2 (link) — one-line description
- Project 3 (link) — one-line description

## Tech Stack
Python | LangChain | OpenAI | ChromaDB | Docker | FastAPI | Streamlit
```

Free deployment platforms:

- **Streamlit Cloud** — Best for AI apps
  https://streamlit.io/cloud
- **Vercel** — Best for Next.js
  https://vercel.com
- **Railway** — Best for APIs
  https://railway.app
- **Render** — Best for Docker
  https://render.com

Goal:

By end of Week 10: 3-5 projects on GitHub with clean READMEs and live demo links.

WEEK 11-12: APPLY BEFORE YOU'RE READY

The truth: You will NEVER feel ready. Apply anyway. The people who break into AI aren't the smartest — they're the ones who started before they felt ready.

Job titles to search for:

- AI Engineer
- LLM Engineer
- ML Engineer (many are actually LLM/AI roles now)
- Prompt Engineer
- AI Developer
- Python Developer (AI/ML)
- Full Stack AI Engineer

Where to find jobs:

- **LinkedIn Jobs** — Set alerts for 'AI Engineer'
  https://linkedin.com/jobs
- **Wellfound (AngelList)** — Startups hiring AI engineers
  https://wellfound.com
- **AI Jobs**
  https://aijobs.net
- **We Work Remotely** — Remote AI roles
  https://weworkremotely.com
- **RemoteOK**
  https://remoteok.com

Daily job hunting routine:

- **Morning:** Apply to 10 jobs (don't overthink it)
- **Afternoon:** Cold DM 5 engineers on LinkedIn (ask for coffee chats)
- **Evening:** Work on one more project or improve existing ones

Cold DM template:

```
Hi [Name], I'm breaking into AI engineering and I really admire
your work at [Company]. Would you have 15 minutes for a virtual
coffee chat? I'd love to learn about your experience.
No pressure either way — thanks for your time!
```

Interview prep — what to know:

- Explain your RAG project end-to-end
- Know the difference between embeddings and fine-tuning
- Understand chunking strategies and why they matter
- Be able to whiteboard a basic LLM application architecture
- Know when to use RAG vs fine-tuning vs prompt engineering

Goal:

By end of Week 12: You've applied to 50+ jobs, had 5+ coffee chats, and landed at least 2-3 interviews.

THE COMPLETE RESOURCE LIST

Free Learning Platforms

Resource	Link	What It's For
Kaggle Learn	https://www.kaggle.com/learn	Python basics
CS50 Python	https://cs50.harvard.edu/python/	Python fundamentals
DeepLearning.AI	https://www.deeplearning.ai/short-courses/	Prompt engineering, LLMs
LangChain Academy	https://academy.langchain.com/	LangChain + RAG
Hugging Face Course	https://huggingface.co/learn	NLP + LLMs
Google ML Crash Course	https://developers.google.com/machine-learning/crash-course	ML basics (later)
fast.ai	https://course.fast.ai/	Deep learning (later)

Essential Tools

Tool	Link	What It Does
Python	https://python.org	The only language you need
Cursor	https://cursor.com	AI-powered code editor
OpenAI API	https://platform.openai.com	GPT-4, embeddings
LangChain	https://langchain.com	LLM app framework
ChromaDB	https://www.trychroma.com	Free vector database
Streamlit	https://streamlit.io	Build UI in Python
GitHub	https://github.com	Your portfolio lives here
Docker	https://docker.com	Containerize your apps

YouTube Channels to Follow

- **Corey Schafer** — Python tutorials → https://www.youtube.com/@coreyms
- **freeCodeCamp** — Everything tech → https://www.youtube.com/@freecodecamp
- **James Briggs** — LangChain, RAG, vectors → https://www.youtube.com/@jamesbriggs
- **Matt Wolfe** — AI news and tools → https://www.youtube.com/@maboroshi
- **Yannic Kilcher** — AI paper explanations → https://www.youtube.com/@YannicKilcher
- **3Blue1Brown** — Math intuition → https://www.youtube.com/@3blue1brown
- **NetworkChuck** — Fun tech content → https://www.youtube.com/@NetworkChuck

FINAL WORD

You don't need a CS degree.
You don't need a $5,000 bootcamp.
You don't need 2 years of preparation.

You need Python, an API key, and 12 weekends of focused building.

The people who break into AI aren't the smartest.
They're the ones who started before they felt ready.

Start today. Build something dumb. Make it better tomorrow.

Follow **@tech_tanajs** on Instagram for daily AI engineering content.